



## RESEARCH ARTICLE OPEN ACCESS

# Prenatal Household Income Instability and Infant Subcortical Brain Volumes

Genevieve Patterson<sup>1</sup> | Alexander J. Dufford<sup>2,3</sup> | Taryn W. Morrissey<sup>4</sup> | Martin Styner<sup>5</sup> | Sun Hyung Kim<sup>5</sup> | Pilyoung Kim<sup>1,6</sup>

<sup>1</sup>Department of Psychology, University of Denver, Denver, Colorado, USA | <sup>2</sup>Center for Mental Health Innovation, Oregon Health & Science University, Portland, Oregon, USA | <sup>3</sup>Department of Psychiatry, Oregon Health & Science University, Portland, Oregon, USA | <sup>4</sup>School of Public Affairs, American University, Washington, District of Columbia, USA | <sup>5</sup>Department of Psychiatry, University of North Carolina Chapel Hill, Chapel Hill, North Carolina, USA | <sup>6</sup>Department of Psychology, Ewha Womans University, Seoul, South Korea

**Correspondence:** Pilyoung Kim ([Pilyoung.Kim@du.edu](mailto:Pilyoung.Kim@du.edu))

**Received:** 16 August 2024 | **Revised:** 12 February 2025 | **Accepted:** 9 April 2025

**Funding:** This work was supported by the National Institute of Health (R01HD090068 and R21DA046556) and NARSAD Independent Investigator Grant.

**Keywords:** amygdala | hippocampus | income instability | infant brain

## ABSTRACT

Prenatal stress exposure may negatively influence the development of the amygdala and hippocampus. Although there is significant income instability during pregnancy, and it can increase stress among pregnant parents, the impact of income instability on brain development is not well understood. The present study examined the association between household income losses during pregnancy and hippocampus and amygdala volumes in early infancy. A total of 63 infants from a prospective longitudinal study of pregnant individuals and their infants completed an MRI during natural sleep. The total number of negative month-to-month earnings shocks (defined as an arc percent change [APC] of  $-25$  or greater) was significantly associated with smaller right hippocampal and right amygdala volumes. This suggests that household income losses during the perinatal period are associated with infant brain structure after birth. These findings provide support for the development of public programs that prioritize financial consistency, especially during sensitive periods like pregnancy.

## 1 | Introduction

Experiences in utero may play an important role in brain development, with growing evidence that prenatal stress is associated with infant brain outcomes after birth (for review see Lautarescu et al. 2020). There are many forms of environmental adversity that can lead to increased stress. Of these, lower socioeconomic status (SES) and greater poverty have been consistently associated with poorer physical health, psychological health, cognitive performance, and academic achievement across development (Evans and Kim 2013, Johnson et al. 2016, Kim et al. 2018). Socioeconomic disadvantage has also been associated with differences in brain

structure and function, especially in regions supporting emotion regulation, learning and memory, cognition, language, and literacy, suggesting neurobiological mechanisms may play a role in these relationships (Evans and Kim 2013, Johnson et al. 2016, Kim et al. 2018, Dufford and Kim 2017, Farah 2017, Hair et al. 2015, Kim et al. 2013, Leijser et al. 2018, Noble et al. 2015). Emerging evidence also demonstrates associations with measures of SES and brain structure as early as infancy including associations between lower SES and smaller neonate cortical and deep gray matter volumes (Betancourt et al. 2016) and reduced white matter, cortical gray matter, and subcortical gray matter volumes and cortical folding at birth (Triplett et al. 2022).

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2025 The Author(s). *Developmental Science* published by John Wiley & Sons Ltd.

## Summary

- Past research has shown that income instability in the United States is increasing and disproportionately impacts low-income households with children.
- The perinatal period specifically is characterized by changes in household income, and prenatal stress has been associated with infant brain development.
- We found that the number of negative month-to-month household earnings shocks across pregnancy was associated with smaller infant right hippocampus and amygdala volumes after birth.

Recent theoretical frameworks for understanding the impact of adversity during development have suggested three dimensions: deprivation, threat, and unpredictability (Berman et al. 2022, McLaughlin et al. 2021, Ellis et al. 2022, Usacheva et al. 2022). Although poverty has often been primarily conceptualized as an example of deprivation (McLaughlin and Sheridan 2016, Cherel et al. 2015), other work has treated poverty as an exposure that increases risk for both deprivation and threat but is not equivalent to either (Berman et al. 2022, McLaughlin et al. 2014). Other theoretical frameworks have focused on harshness and unpredictability as the key environmental dimensions influencing development (Belsky et al. 2012). Within these frameworks income-to-needs ratio (INR) is often conceptualized as harshness, while residential, household, and parental job changes are conceptualized as unpredictability (Belsky et al. 2012). Research with both human subjects and animal models has illustrated the relationship between the dimension of unpredictability and stress response. Within animal model research, unpredictability has been consistently utilized to elicit behavioral and physiological stress response, with chronic unpredictable mild stress protocols producing anxiety and depression like behavior in rats (Hu et al. 2017) and increased serum corticosterone in mice (Monteiro et al. 2015). Environmental unpredictability also predicts stronger glucocorticoid stress responses in tree swallows (Zimmer et al. 2020). In experimental research, participants receiving unpredictable shocks report greater anxiety than those receiving predictable shocks (Havranek et al. 2016). Additionally, early life unpredictability has been associated with higher emotional and behavioral problems in adolescence, moderated by stress reactivity (Buthmann et al. 2025). Despite more nuanced research into the effects of SES on development, there is still limited research on the role of unpredictability in income and its unique contributions to stress and development, especially in the perinatal period.

Income instability, or the number of substantial changes in income over a given period, is becoming increasingly common in the United States and disproportionately impacts low-income households. Research using nationally representative samples indicates that income instability doubled from 1980 to 2009; in any specific year since 1996, income instability among families in the bottom 10% of the income distribution exceeded the top 1% (Hardy and Ziliak 2014). Between 1991/1992 and 2002/2003, low-income households experienced the greatest monthly income volatility and increases in volatility (Bania and Leete 2009).

Although social safety net programs are intended to reduce household income instability, since 1980, these programs have become less effective at addressing instability for those with the lowest incomes (Hardy 2017). Overall, low-income families experience economic insecurity and precarity (for review, see Cooper and Pugh 2020). Like poverty, income instability is especially common among households with young children. For example, Fonov et al. (2011) found that large family income losses increased more than large family income gains for children in low-income households. The perinatal period is characterized by income and other resource changes (Heflin et al. 2024, Stanczyk 2020). Qualitative research shows that low-income women report financial strain and employment changes attributed to pregnancy (Gross et al. 2019). There is currently no federal policy in the United States providing or guaranteeing access to paid family leave and state-level policies vary. Although state-paid family leave programs help smooth household resources (Stanczyk 2019), most states lack paid family leave laws, and public assistance program benefits do not offset losses in earnings surrounding childbirth (Heflin et al. 2024).

Independent of average income, income instability is associated with poorer physical, mental, and behavioral health (e.g., Hill et al. 2013). Income volatility has been associated with increased cardiovascular disease risk and all-cause mortality in adulthood (Elfassy et al. 2019), increased depression in young adults and teen parenthood (Whitfield et al. 2021), increased early childhood behavior problems (Grasset et al. 2019), worse school attendance in children (Gennetian et al. 2018), higher rates of food insecurity among households with children (Haile et al. 2001), and increased psychiatric disorders risk in early adulthood (Cheng et al. 2020). Income losses may be especially impactful given negative impacts on household resources and stress. Indeed, recent research found family income losses predict both externalizing and internalizing symptoms throughout development (Miller et al. 2021). Income loss frequency was also associated with depressive symptoms in adulthood when absolute income volatility was high (Prause et al. 2009). Although research on prenatal income instability is limited, a systematic review found other forms of instability such as housing instability are associated with numerous adverse pregnancy outcomes (DiTosto et al. 2021).

Emerging evidence indicates that income instability specifically is associated with brain health. Higher income volatility and income drops over a 20-year period were associated with poorer performance on processing speed and executive functioning tasks and worse microstructural integrity (Grasset et al. 2019). However, to our knowledge, no studies have investigated the relationship between income instability and children's brain structure, particularly in infancy.

Household income instability during the perinatal period may affect the developing fetal brain through prenatal stress. Research shows the amygdala and hippocampus are sensitive to stress exposure, partially due to the many glucocorticoid receptors present in these regions (Aronsson et al. 1988, Badihian et al. 2020, McEwen et al. 1992). Animal model research has illuminated potential mechanisms through which perinatal environmental stressors can impact amygdala and hippocampus development (for review, see Charil et al. 2010). For

example, rats exposed to stress in utero have significantly altered hippocampal gene expression (Bogoch et al. 2007) and amygdala excitatory/inhibitory imbalance (Marchisella et al. 2021).

Prenatal stress is associated with the amygdala and hippocampus structure in humans. For instance, exposure to stressful life events prenatally predicted smaller amygdala nuclei in young adulthood (Mareckova et al. 2022). Prenatal disaster-related stress has been associated with larger bilateral amygdala volumes in childhood (Jones et al. 2019) and larger bilateral amygdala and hippocampus in young adulthood (Li et al. 2023). Moog et al. (2021) found prenatal self-reported perceived stress was associated with smaller neonate left hippocampal volume. Prenatal anxiety symptoms have been associated with smaller bilateral hippocampal growth over the first 6 months (Qiu et al. 2013). Overall, there are inconsistencies in the direction of these associations (larger vs. smaller volumes). This may be partly due to heterogeneous operationalizations of prenatal stress and assessment of volume at different ages. In addition to brain structure, prenatal stress has also been associated with hippocampus and amygdala functional connectivity (Humphreys et al. 2020, Scheinost et al. 2016, Scheinost et al. 2020) and diffusion tensor imaging metrics (Humphreys et al. 2020, Lautarescu et al. 2020, Rifkin-Graboi et al. 2013) in infants.

SES is also associated with amygdala and hippocampal volumes in youth, with consistent associations between lower SES and smaller hippocampal volumes, with mixed results for amygdala volume (Rakesh and Whittle 2021). In early life, there is evidence that prenatal SES is associated with infant brain structure (Triplett et al. 2022, Spann et al. 2020), infant functional connectivity (Ramphal et al. 2020), and fetal brain structure (Lu et al. 2021).

Despite growing evidence that prenatal stressors, including low income and social disadvantage during pregnancy, can influence infant brain outcomes, little is known about prenatal income instability. Familial SES reflects a complex environmental exposure that includes multiple dimensions. Among these dimensions, unpredictability in income is particularly understudied. Given that income instability is becoming more prevalent (Hardy and Ziliak 2014) and has been associated with poor physical, mental health, and behavioral outcomes independent of average income (Hill et al. 2013), it is important to understand the role of income dynamics throughout development. The perinatal period specifically is a time of relatively dramatic income changes (Stanczyk 2019) and, therefore, heightened vulnerability to income instability. Thus, perinatal income instability and infant brain development research can inform public policies that best support two generations during this critical period. Poverty reduction programs often do not account for the cognitive burden of income shocks, which may impact behaviors such as program enrollment and utilization, or directly address variability in income resulting from uncertain wages and hours (Gennetian and Shafir 2015). Therefore, a better understanding of the role of instability, specifically, can help maximize the effectiveness of program design. Here, we investigated the relationship between income instability during pregnancy, specifically household earnings losses, and infant amygdala and hippocampus volumes. We chose these regions due to susceptibility to stress exposure and focused on earnings losses, as they may particularly affect

household resources and stress. We hypothesized that more income losses from conception to birth would be associated with reduced infant hippocampal volumes after birth and differences in infant amygdala volume after birth. However, given inconsistencies in the literature, we did not hypothesize a direction for the amygdala. We also examined the effect of prenatal INR on infant amygdala and hippocampus volume, allowing us to compare average income with instability in association with brain outcomes.

## 2 | Methods

### 2.1 | Participants

The study sample consisted of 63 birthing parents and their infants ( $M_{\text{age at scan}} = 4.98 \text{ weeks} \pm 2.85$ ; 52.4% female). Study recruitment focused on low- and middle-income pregnant participants ( $M_{\text{prenatal income-to-needs ratio}} = 3.71 \pm 2.60$ ), and 31.7% of the sample was low income (average familial prenatal income to needs ratio [INR]  $\leq 2$ , see Section 2.3.2 below for calculation of INR). See Table 1 for demographic characteristics of the sample. To estimate the sample size for our multiple regressions, we performed a power analysis with G\*Power, 3.1 (Faul et al. 2007) to test  $R^2$  increase for a model with four predictors and one tested predictor. The test revealed a sample size of 55 to detect a medium effect size ( $f^2 = 0.15$ ) with 80% power at a 0.05 alpha level. This effect size estimate was chosen based on previously reported medium effect sizes (Cohen's  $f^2 = 0.27$ , 0.29) for association between stress exposure during pregnancy and amygdala volumes (Mareckova et al. 2022) and differences in whole hippocampus and amygdala volumes for participants exposed to maternal stress prenatally (partial eta squared  $> 0.06$ , Li et al. 2023). In addition, medium effects were found for the correlations between SES and gray matter volumes in 1-month-old infants (deep gray  $r = 0.34$ ; cortical gray  $r = 0.38$ ) and Cohen's  $f^2$  for the local effect of the SES composite computed from reported regression models was between 0.10 and 0.15 (Betancourt et al. 2016). Thus, we used an estimated  $f^2 = 0.15$  for our power analysis.

Study participants were recruited during the first trimester of pregnancy from obstetrics and gynecology clinics in the Denver metro area by an on-site research coordinator who screened prenatal patients for eligibility based on medical records and spoke to potential participants at routine prenatal visits. Eligibility criteria included age between 18 and 45 years, singleton intrauterine pregnancy prior to 14 weeks of gestation, fluency in English or Spanish, and an INR below 8 based on the phone screening for the recruitment. Eligibility criteria for INR were based on self-reported income during the screening call. During more detailed interviews conducted at the first prenatal visit, four participants reported INR greater than 8. Data were collected from March 2018 to June 2022. Due to changes in response to the COVID-19 pandemic restrictions, two of the participants started the current study in the third trimester.

MRI data acquisition was attempted for 96 infants. Twenty infants failed to sleep through the MRI sequences. Following visual inspection of T1w and T2w images, 13 infants were excluded for high motion in either the T1w or T2w scan. Of the 63 infants

**TABLE 1** | Demographic characteristics.

| <b>Birth parent characteristics</b>                                                       | <b>(<i>M</i> ± <i>SD</i>) or %</b> |
|-------------------------------------------------------------------------------------------|------------------------------------|
| Average prenatal EPDS ( <i>M</i> ± <i>SD</i> )                                            | 5.9 ± 4.3                          |
| Average prenatal STAI-S ( <i>M</i> ± <i>SD</i> )                                          | 29.9 ± 10.6                        |
| Race (%)                                                                                  |                                    |
| American Indian and/or Alaskan Native                                                     | 3.2                                |
| Black or African American                                                                 | 9.5                                |
| Native Hawaiian or other Pacific Islander                                                 | 3.2                                |
| White                                                                                     | 71.4                               |
| Other <sup>a</sup>                                                                        | 12.7                               |
| Ethnicity (%)                                                                             |                                    |
| Not Hispanic/Latino                                                                       | 68.3                               |
| Hispanic/Latino                                                                           | 31.7                               |
| Average prenatal income-to-needs ratio ( <i>M</i> ± <i>SD</i> )                           | 3.7 ± 2.6                          |
| Age in years ( <i>M</i> ± <i>SD</i> )                                                     | 29.1 ± 5.8                         |
| Relationship status (%)                                                                   |                                    |
| Single                                                                                    | 7.9                                |
| Dating                                                                                    | 11.1                               |
| Engaged                                                                                   | 6.3                                |
| Married                                                                                   | 58.7                               |
| Long-term relationship                                                                    | 6.3                                |
| Common law marriage                                                                       | 6.3                                |
| Other <sup>b</sup>                                                                        | 3.2                                |
| Prenatal nicotine or substance use <sup>c</sup> (% positive immunoassay)                  | 25.4                               |
| Education, highest degree completed (%)                                                   |                                    |
| Less than high school                                                                     | 6.3                                |
| High school diploma/GED                                                                   | 41.3                               |
| Associate degree                                                                          | 11.1                               |
| Bachelor's degree                                                                         | 19                                 |
| Master's degree                                                                           | 19                                 |
| MD/PhD                                                                                    | 3.2                                |
| <b>Infant characteristics</b>                                                             |                                    |
| Age at scan in weeks ( <i>M</i> ± <i>SD</i> )                                             | 4.9 ± 2.8                          |
| Gestational age at birth in weeks ( <i>M</i> ± <i>SD</i> )                                | 38.9 ± 1.7                         |
| Postconceptional age at scan in weeks ( <i>M</i> ± <i>SD</i> )                            | 43.8 ± 2.5                         |
| Birthweight in pounds ( <i>M</i> ± <i>SD</i> )/birthweight in kg ( <i>M</i> ± <i>SD</i> ) | 7.0 ± 1.2/3.1 ± 0.5                |
| Sex assigned at birth (%)                                                                 |                                    |
| Female                                                                                    | 52.4                               |
| Male                                                                                      | 47.6                               |
| NICU admission (% admitted)                                                               | 19                                 |
| Race (%)                                                                                  |                                    |
| American Indian and/or Alaskan native                                                     | 1.6                                |
| Black or African American                                                                 | 7.9                                |
| White                                                                                     | 60.3                               |
| Other                                                                                     | 30.2                               |
| Ethnicity (%)                                                                             |                                    |
| Not Hispanic/Latino                                                                       | 66.7                               |
| Hispanic/Latino                                                                           | 33.3                               |

<sup>a</sup>Four parents identified their race as Hispanic, Latina, or Mexican American and four parents identified as multiracial. Of those who identified as multiracial one identified as American Indian, Black, and White, one identified as Filipino and Latino, one identified as African American, White, and Native American, and one identified as Peruvian and White.

<sup>b</sup>One parent reported "divorced," one did not report relationship status.

<sup>c</sup>Three parents tested positive for nicotine, 12 tested positive for cannabis, and 1 tested positive for morphine.



who were included in the analysis, 27 overlap with (Kim et al. 2021), which performed a resting-state functional connectivity analysis.

## 2.2 | Procedure

The research study included three prenatal home visits (conducted at approximately 12-, 24-, and 32-weeks' gestation) and two postnatal home visits (conducted at approximately 1 month and 1 year after delivery). At each home visit, participants participated in interviews, including assessment of household income (see "Income Shocks" below). In response to the COVID-19 pandemic, home visits occurring after 03/13/2020 transitioned to remote visits or lab visits based upon university regulations at the time of the visit. Interviews were conducted via video conferencing for visits conducted remotely. An MRI scan for birthing parents and infants was also conducted at approximately 1 month after delivery (see "MRI acquisition" below). All participants provided informed written consent as approved by the University of Denver Institutional Review Board.

## 2.3 | Measures

### 2.3.1 | Demographics

Birthing parent age, educational attainment, and relationship status were collected via interview at the first prenatal visit. Race and ethnicity for birthing and non-birthing parents were collected at birthing parent interviews at the first prenatal visit and infant race and ethnicity were collected at the first postnatal visit. Infant sex, gestational age at birth, birthweight, and a binary indicator of infant NICU admission was collected from medical records. For 15 participants, medical record information was not available; therefore, information collected at the birthing parent interview at the first postnatal visit was used. Participants provided a urine sample for a Nic-Alert test to evaluate for nicotine metabolite and a CLIA-waived 5-panel drug immunoassay at their first visit. Prenatal exposure to substances or nicotine was defined as a positive immunoassay. Birthing parent anxiety and depression symptoms were assessed at each prenatal visit using the state anxiety scales from the State-Trait Anxiety Inventory (STAI-S; Spielberger et al. 1983) and the Edinburgh Postnatal Depression Scale (EPDS; Cox et al. 1987). Average prenatal anxiety and depression symptoms were each calculated by averaging STAI-S and EPDS scores respectively across prenatal visits with higher scores indicating more symptoms.

### 2.3.2 | Income Instability & INR

At each prenatal visit, birthing parents were interviewed about all forms of household income including occupation, hours, and earnings for all adults in the household as well as any other sources of income such as rent from properties, child support, social security, disability or veterans' benefits, unemployment benefits, workman's compensation, help from relatives, scholarships. For all sources of income, birthing parents were asked to provide the date the job started or ended, as well as dates for

any changes in pay rate or changes in hours. Dates of significant life events (e.g., birthdays, holidays) were used to support recall. This calendar interview methodology has been previously used for retrospective reports of income (De France et al. 2022, Evans et al. 2021, Kim et al. 2020, Marcynyszyn et al. 2008) and has been shown to provide accurate and reliable data (Belli 1998, Belli et al. 2004, Glasner and van der Vaart 2009). All participants provided information about sources of household income for the full prenatal period, and 46 participants provided income information for the months immediately prior to the child's birth. From this interview, household earned income was calculated for each month from the month prior to conception until the month of the child's birth. Government support in the form of benefits from temporary assistance to needy families (TANF; or cash welfare), the Supplemental Nutrition Assistance Program (SNAP, formerly known as food stamps, with benefits distributed as vouchers that can be used to purchase foods to be prepared at home), and the SNAP for women, infants, and children (WIC) was excluded from the calculation of household earned income.

**2.3.2.1 | Prenatal INR.** INR was calculated by dividing the total reported household earned income from all household members by the poverty threshold corresponding to the number of persons and dependents living in the household during the prenatal period, specified by the US Census Bureau for that year. Since the poverty threshold is based upon an annual income value, annual income was extrapolated for each month from conception to birth based on the participants' reported income for that month, allowing for an INR calculation for each month. Average prenatal INR was computed by averaging INR for each month from conception to the month of the child's birth. The study sample had a mean INR of 3.70 with a standard deviation of 2.60 and a range from 0.28 to 11.62. See Figure S1 for the distribution of average prenatal INR in the sample.

**2.3.2.2 | Prenatal Income Instability.** The arc percent change (APC) in monthly household earned income was calculated as a measure of income instability. APC is a percent change measure bounded by  $-200$  to  $200$ , with higher absolute values indicating greater instability. The APC allows for the presence of zero values (thereby allowing for participants who had months with zero earnings) and is less dependent on average income and has been used in previous research on income volatility (Hardy and Ziliak 2014, Gennetian et al. 2018, Morrissey et al. 2020). An income shock was defined as  $APC > \pm 25$  for a given change in earnings from 1 month to the next. This threshold has been used previously in research, and income shocks of this magnitude have been associated with child and adult outcomes (Miller et al. 2021, Hardy 2014). The total prenatal negative income shocks were then calculated by summing the number of negative shocks across the time from conception to birth ( $M = 0.64$ ,  $SD = 0.87$ ,  $Range = 0-3$ ). The number of positive income shocks was also summed across the time from conception to birth ( $M = 0.59$ ,  $SD = 0.82$ ,  $Range = 0-3$ ). See Figure S2 for the distribution of the number of negative and positive income shocks in the sample. Seventeen participants did not have available monthly income information for the month

prior to conception, and therefore did not have shock information for the month of conception to include in the summation.

### 2.3.3 | MRI Acquisition

Infants were scanned during natural sleep on a Siemens 3T magnet scanner with a 32-channel head coil at the Intermountain Neuroimaging Consortium at the University of Colorado Boulder. Prior to scanning, infants were fed, swaddled, and placed into the scanner. Infants wore wax earplugs and foam cushions were placed around their heads within the head coil to prevent wakefulness from scanner noise and limit head motion. A trained researcher remained lying on the scanner bed with the infant for the duration of the scan to monitor the infant and the birthing parent remained in the scan room if they chose to do so. For scans conducted from March 2020 to June 2022, due to COVID-19 related social distancing rules, the trained researcher monitored the infant from outside of the scanner and a parent remained in the scan room for the duration of the scan.

T1-weighted (T1w) images were acquired using a three-dimensional magnetization-prepared rapid gradient echo sequence (repetition time = 2500 ms; echo time = 2.05 ms; inversion time = 1000 ms; flip angle 8°; 3 min 29 s) and T2-weighted (T2w) images were acquired using a 3D fast turbo spin echo sequence (repetition time = 3200 ms; echo time = 564 ms; flip angle var; 2 min 0 s). The spatial resolution was a 1.0 mm × 1.0 mm × 1.0 mm voxel for both T1w and T2w.

## 2.4 | MRI Image Processing

The T1w and T2w brain images underwent a correction process for intensity non-uniformity employing the N4 bias field correction algorithm (Tustison et al. 2010). The corrected images were then rigidly transformed and aligned with a pediatric neonate atlas in a standardized stereotaxic space, as delineated by Fonov et al. (2011). A brain masking procedure was executed utilizing a majority voting strategy. This involved the joint warping registration of T1w and T2w images with seven predefined single atlases, subsequently ensuring the creation of accurate brain masks. Necessary manual corrections to the brain masks were conducted using the itkSNAP software (Qiu et al. 2013) to guarantee anatomical accuracy.

For the segmentation process, a multi-modality (T1w and T2w) multi-atlas segmentation workflow was employed, utilizing the in-house, open-source MultiSegPipeline software (Cherel et al. 2015). Whole brain tissue labels were segmented as white matter, gray matter, and cerebrospinal fluid compartments. Concurrently, a region of interest segmentation was carried out, encompassing the hippocampus and amygdala regions (Moog et al. 2018, Overfeld et al. 2020; see Figure 1 for visualization of regions of interest). The MultiSegPipeline software employs the label fusion method from the advanced normalization tools (ANTs) toolset (Avants et al. 2009, Tustison et al. 2021). An extensive visual assessment was conducted to ascertain the segmentation quality of all images in terms of anatomical accuracy, leading to the conclusion that no processed data warranted exclusion based on the segmentation quality.

## 2.5 | Analyses

### 2.5.1 | Evaluation of Covariates

Bivariate correlations were conducted to test the following as potential covariates: total intracranial volume (ICV), postconceptional age at scan (sum of gestational age at birth and age at scan), sex at birth, birthweight, average prenatal INR, presence of a NICU stay, and prenatal exposure to nicotine or recreational substances. Variables were included as covariates if they were significantly associated with hippocampal or amygdala volumes at  $p < 0.05$ , therefore, ICV, postconceptional age at scan, and birthweight were included in further analyses.

### 2.5.2 | Analyses

Separate multiple regression models were run to predict infant right and left hippocampus and amygdala volume from the total number of negative income shocks, controlling for ICV, postconceptional age at scan, and birthweight. Bonferroni correction across the four brain regions was applied ( $p = 0.0125$ ). Next, we ran analysis with INR predicting infant right and left hippocampus and amygdala volume, controlling for ICV, postconceptional age at scan, and birthweight to examine the effects of overall income levels. Given the correlation between the total negative shocks and prenatal substance exposure, we ran additional analyses predicting brain volumes from prenatal substance exposure. Although we specifically were interested in the role of earnings losses as a potential stressor, analyses predicting brain volume from positive income shocks were also conducted (see Supporting Materials).

## 3 | Results

### 3.1 | Demographic Variables and Correlations

Within our sample, 43% of participants experienced at least one negative income shock from conception to birth month (see Figure S2 for the distribution of negative income shocks). Lower average prenatal INR was significantly associated with a greater number of negative income shocks (see Table 2).

Sex at birth, average prenatal INR, NICU stay, and prenatal exposure to nicotine or recreational substances were not significantly correlated with infant hippocampal or amygdala volumes (see Table 2). ICV, birthweight, and postconceptional age at scan were significantly associated with amygdala and/or hippocampal volumes (see Table 2) and therefore were included as covariates in all analyses.

The number of negative income shocks was not significantly correlated with ICV, birthweight, or postconceptional age at scan (see Table 2).

Average birthing parent prenatal anxiety symptoms were correlated with the number of negative income shocks ( $r(61) = 0.38$ ,  $p = 0.002$ ) and right hippocampus ( $r(61) = -0.30$ ,  $p = 0.016$ ) and right amygdala volumes ( $r(61) = -0.26$ ,  $p = 0.043$ ). Prenatal anxiety symptoms were not correlated with the number

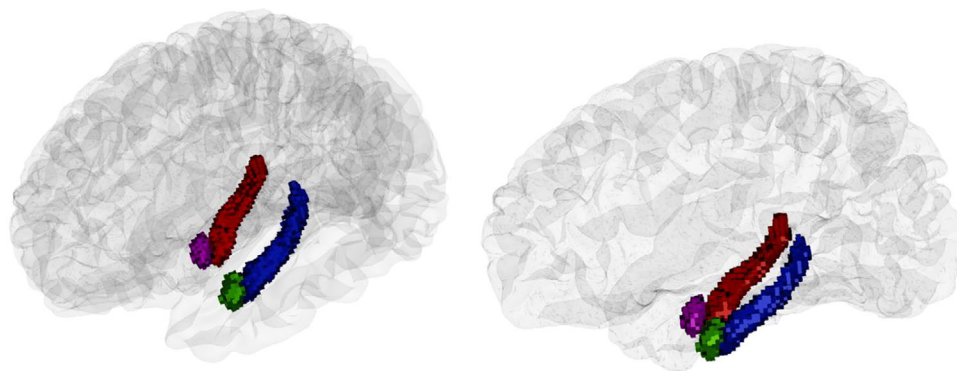
**TABLE 2** | Correlation matrix of study variables.

|                 | Right Hipp. | Left Hipp. | Right Amy. | Left Amy. | Sum Neg. shocks | Sum Pos. shocks | ICV    | PC age at scan | Infant sex | BW    | INR     | NICU stay | Sub. Exp. | STAI-S | EPDS |
|-----------------|-------------|------------|------------|-----------|-----------------|-----------------|--------|----------------|------------|-------|---------|-----------|-----------|--------|------|
| Right Hipp.     | —           |            |            |           |                 |                 |        |                |            |       |         |           |           |        |      |
| Left Hipp.      | 0.71**      | —          |            |           |                 |                 |        |                |            |       |         |           |           |        |      |
| Right Amy.      | 0.46**      | 0.38**     | —          |           |                 |                 |        |                |            |       |         |           |           |        |      |
| Left Amy.       | 0.39**      | 0.42**     | 0.70**     | —         |                 |                 |        |                |            |       |         |           |           |        |      |
| Sum Neg. shocks | −0.25*      | −0.20      | −0.32*     | −0.17     | —               |                 |        |                |            |       |         |           |           |        |      |
| Sum Pos. shocks | −0.14       | −0.06      | 0.02       | 0.18      | 0.38**          | —               |        |                |            |       |         |           |           |        |      |
| ICV             | 0.47**      | 0.35**     | 0.44**     | 0.40**    | −0.02           | −0.02           | —      |                |            |       |         |           |           |        |      |
| PC age at scan  | 0.24        | 0.03       | 0.37**     | 0.25*     | −0.09           | −0.03           | 0.72** | —              |            |       |         |           |           |        |      |
| Infant sex      | −0.13       | −0.11      | 0.01       | −0.12     | −0.15           | −0.13           | −0.27* | −0.17          | —          |       |         |           |           |        |      |
| BW              | 0.30*       | 0.25*      | 0.02       | 0.10      | 0.06            | −0.12           | 0.06   | −0.03          | 0.06       | —     |         |           |           |        |      |
| INR             | 0.01        | 0.17       | 0.21       | 0.21      | −0.46**         | −0.31*          | −0.05  | −0.07          | 0.08       | −0.02 | —       |           |           |        |      |
| NICU stay       | −0.16       | −0.14      | −0.13      | −0.24     | 0.02            | 0.10            | 0.14   | 0.14           | −0.10      | −0.23 | −0.23   | —         |           |        |      |
| Sub. Exp.       | −0.06       | 0.03       | −0.13      | −0.06     | 0.38**          | 0.12            | 0.05   | −0.06          | 0.12       | 0.22  | −0.33** | 0.09      | —         |        |      |
| STAI-S          | −0.30*      | −0.22      | −0.26*     | −0.13     | 0.38**          | 0.12            | −0.16  | −0.15          | 0.11       | −0.10 | −0.24   | 0.17      | 0.30*     | —      |      |
| EPDS            | −0.39**     | −0.24      | −0.26*     | −0.22     | 0.38**          | −0.02           | −0.21  | −0.18          | 0.14       | −0.03 | −0.14   | 0.22      | 0.24      | 0.79** | —    |

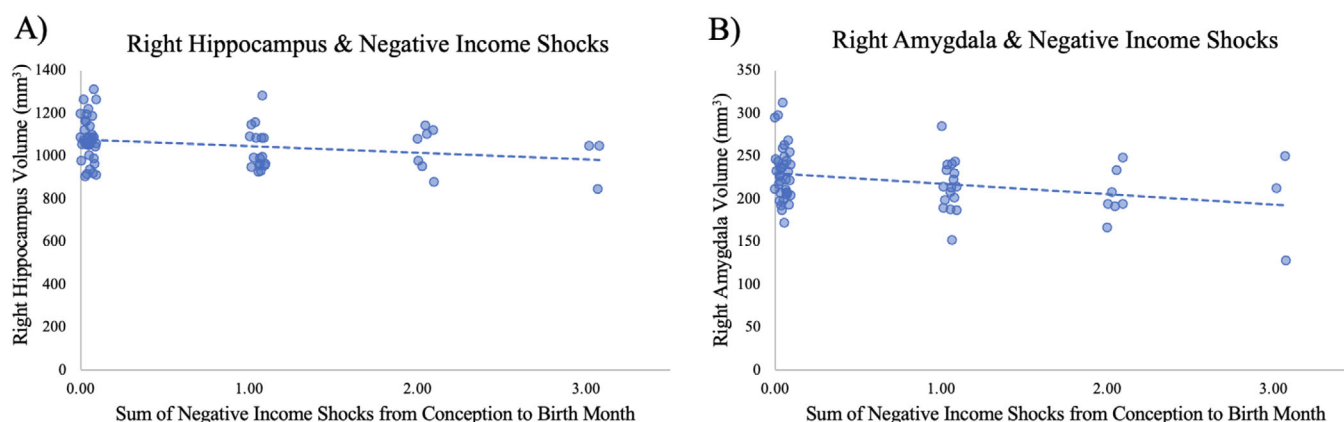
Abbreviations: Amy. = amygdala, BW = birthweight, EPDS = Edinburgh postnatal depression scale score, Hipp. = hippocampus, ICV = intercranial volume, INR = average prenatal volume, INR = average prenatal income to needs ratio, Neg. = negative, PC = postconceptional, Pos. = positive, STAI-S = average prenatal State Trait Anxiety Inventory State score, Sub. Exp. = prenatal substance exposure.

\* $p < 0.05$ .

\*\* $p < 0.01$ .



**FIGURE 1** | Regions of interest, 3D visualization of hippocampus and amygdala segmentation. Blue = left hippocampus, Green = left amygdala, Red = right hippocampus, Purple = right amygdala.



**FIGURE 2** | Scatter plots for sum of negative income shocks and infant right hippocampus and amygdala volume. (A) Scatter plot for sum of negative income shocks and infant right hippocampus volume (B) Scatter plot for sum of negative income shocks and infant right amygdala volume.

of positive income shocks ( $r(61) = 0.11$ ,  $p = 0.376$ ) or with average prenatal INR ( $r(61) = -0.24$ ,  $p = 0.062$ ).

Average birthing parent prenatal depressive symptoms were also correlated with the number of negative shocks ( $r(61) = 0.38$ ,  $p = 0.002$ ) and right hippocampus ( $r(61) = -0.39$ ,  $p = 0.001$ ) and right amygdala volumes ( $r(61) = -0.26$ ,  $p = 0.042$ ). Prenatal depressive symptoms were also not correlated with the number of positive income shocks ( $r(61) = -0.02$ ,  $p = 0.859$ ) or with average prenatal INR ( $r(61) = -0.14$ ,  $p = 0.269$ ).

### 3.2 | Associations Between Income Shocks/Average Income and Infant Brain Structure

Turning to the regression models, the number of negative income shocks was significantly associated with smaller right hippocampal volume ( $\beta = -0.27$ ,  $p = 0.012$ ), after controlling for ICV, postconceptional age at scan, and birthweight (see Figure 2 for scatterplot of association with raw data). This association remained significant after applying the Bonferroni correction across the four brain regions ( $p < 0.0125$ ). In addition, the number of negative income shocks was significantly associated with left hippocampal volume, controlling for ICV, postconceptional age at scan, and birthweight ( $\beta = -0.23$ ,  $p = 0.038$ ), but was no longer statistically significant after the Bonferroni correction.

Further, the number of negative income shocks was significantly associated with smaller right amygdala volume ( $\beta = -0.30$ ,  $p = 0.009$ ) in models with covariates, and remained significant with the Bonferroni correction across the four brain regions ( $p < 0.0125$ ; see Figure 2 for scatterplot of association with raw data). However, the number of negative income shocks was not significantly associated with left amygdala volume ( $\beta = -0.17$ ,  $p = 0.16$ ) in regression models.

Cohen's  $f^2$  for local effect size was calculated as a measure of local effect size for negative income shocks within the regression models. The effect size was small for both right hippocampus ( $f^2 = 0.12$ ) and right amygdala ( $f^2 = 0.13$ ) volumes.

Average prenatal INR (a measure of average earnings) was associated with right amygdala volume, controlling for ICV, postconceptional age at scan, and birthweight; however, this did not survive correction ( $\beta = 0.24$ ,  $p = 0.039$ ). Average prenatal INR was not significantly associated with left amygdala volume ( $\beta = 0.23$ ,  $p = 0.052$ ) or right or left hippocampal volumes (right:  $\beta = 0.03$ ,  $p = 0.78$ ; left:  $\beta = 0.18$ ,  $p = 0.11$ ) in regression models with controls.

Prenatal substance exposure was not associated with right or left amygdala or hippocampal volumes covarying for ICV,



postconceptional age at scan, and birthweight (all  $\beta$ s between  $-0.17$  and  $-0.08$ , all  $p$  values  $> 0.14$ ).

The number of positive income shocks was unassociated with right or left amygdala or hippocampal volumes after controlling for ICV, postconceptional age at scan, and birthweight (see Supporting Materials).

An additional sensitivity analysis was run, excluding participants who were missing preconception income information and, therefore, income shock data for the month of conception. The pattern of results remained consistent (see Supporting Materials).

In order to test whether the lateralized pattern of findings could be attributed to differences in segmentation quality between the hemispheres, additional analysis was run to test differences in data quality between the hemispheres. For each subject, a segmentation quality score between 1 and 4 was assigned for both left and right amygdala and hippocampus during visual data quality checks. Quality scores were averaged for right hemisphere and left hemisphere structures, and a paired samples  $t$ -test was performed to test differences in data quality between the hemispheres. There were no significant differences in segmentation quality between the right hemisphere ( $M = 3.40$ ,  $SD = 0.48$ ) and left hemisphere structures ( $M = 3.40$ ,  $SD = 0.43$ ,  $t[62] = -0.27$ ,  $p = 0.78$ ).

## 4 | Conclusions

Consistent with previous research findings that, on average, United States households experience declines in income in the time around a birth (Stanczyk 2020), we found that 43% of participants in our sample experienced at least one negative income shock in the time period assessed, with some participants experiencing up to three shocks. Due to our targeted recruitment of pregnant individuals with low and middle incomes, our sample had a relatively higher representation of participants with low income (31.7%), compared to the 2020 US Census report that 27.5% of the population had family incomes less than 200% of the poverty threshold (Shrider et al. n.d.). Consistent with prior evidence that lower-income families are most susceptible to income instability (Hardy and Ziliak 2014, Cooper and Pugh 2020, Fonov et al. 2011), we found that lower INR was associated with a higher number of negative income shocks. However, participants across the income distribution in our sample experienced income losses, as is consistent with prior research focused upon the perinatal period (Stanczyk 2020). The distribution of income shocks in our sample was similar to other studies of income instability (Elfassy et al. 2019, Whitfield et al. 2021, Miller et al. 2021, Grasset et al. 2019); however, many currently published studies rely upon data collected quarterly or annually over many years so the current research fills a needed gap in looking at instability over a shorter timescale during a critical developmental period.

We found that prenatal income losses are associated with smaller right hippocampus and amygdala volumes in infants after birth. However, average prenatal INR—reflecting average household resources—was not associated with these brain volumes. This may suggest that income losses, compared to INR, are more sensitive in predicting infant brain outcomes. Earnings losses, not

gains, were associated with brain structure. The earning losses may be associated with reduced resource levels and related stress. Given that prenatal earnings losses are common (Stanczyk 2020), results have implications for research and policy.

These results build upon a growing literature demonstrating associations between prenatal stressors and brain development (Betancourt et al. 2016, Triplett et al. 2022, Spann et al. 2020, Ramphal et al. 2020, Kim et al. 2021, Gao et al. 2015). Our findings are consistent with previous research findings that prenatal stress is associated with smaller hippocampal volumes and reduced hippocampal growth in infancy (Moog et al. 2021, Qiu et al. 2013).

Likewise, findings are consistent with prior work demonstrating associations between prenatal stress and amygdala volumes (Mareckova et al. 2022, Jones et al. 2019, Li et al. 2023, Buss et al. 2012). However, directionality has been mixed, potentially due to heterogenous study design. There is limited research in infancy specifically, with some studies failing to find an association (Moog et al. 2021). However, infant studies using other brain measures have found associations between prenatal stress and amygdala white matter structure and both structural and functional connectivity with the amygdala (Humphreys et al. 2020, Scheinost et al. 2016, Rifkin-Graboi et al. 2013).

Although we did not hypothesize that findings would be lateralized, we found associations with income losses only in the right hippocampus and amygdala. Previous stress research has also reported findings in right lateralized regions in infancy (Rifkin-Graboi et al. 2013, Posner et al. 2016) and childhood (Buss et al. 2012, Wen et al. 2017). There is also evidence from rodent models that glucocorticoid exposure resulting from stress may have more pronounced effects on the right hippocampus (Zach et al. 2010). However, some studies report bilateral associations (Li et al. 2023, Qiu et al. 2013) and others report findings in the left hemisphere (Moog et al. 2021). Given mixed results, further research is needed to clarify these relationships.

Income losses may be uniquely associated with stress in the birthing parent as instability may elicit uncertainty. Indeed, non-human animal studies have shown that unpredictability is a core factor in eliciting stress responses in female dams, indicating that it is the unpredictable, not the predictable, exposure to stressors that leads to long-term behavioral and biological changes in offspring (Freide and Weinstock 1984, Haile et al. 2001). Financial instability is also associated with increased cognitive load and changes in decision-making behaviors (Gennetian and Shafir 2015). Monthly income volatility specifically has been associated with financial planning decisions and lower locus of control (Peetz et al. 2021). Therefore, psychological and behavioral changes following income loss, may have downstream impacts on stress and health via changes in material resources, relationship functioning, and psychological distress. We found that the number of income losses, not gains, was positively associated with birthing parent self-reported anxiety and depressive symptoms, which were also associated with smaller infant brain volumes. Given our study design we were not able to test mediation formally, however these findings suggest a potential mechanism to test in future research. Parental HPA axis functioning is one route by which stress during pregnancy may influence fetal brain development (Lautarescu et al. 2020). Prenatal stress has also

been associated with cortisol metabolism in the placental barrier (O'Donnell et al. 2009), inflammation (Mancuso et al. 2024, Roshan-Milani et al. 2021), and gut microbiota (Frankiensztajn et al. 2020). The hippocampus and amygdala are sensitive to stress exposure (Aronsson et al. 1988, Badihian et al. 2020, McEwen et al. 1992), and therefore may be uniquely vulnerable to stress associated with income losses. However, SES has also been associated with cortical brain volumes and functional connectivity in cortical regions in infancy (Betancourt et al. 2016, Triplett et al. 2022, Ramphal et al. 2020, Gao et al. 2015), therefore future research is needed to explore if income instability is associated with structure and function of other brain regions.

Our study has several limitations. First, while we draw meaningful and prospective associations between prenatal experiences and later brain structures, the design cannot support causality. Second, our sample size was reduced by infants unable to sleep through both T1w and T2w imaging sequences. Therefore, further work with a larger sample size should aim to replicate these findings. Given our focus on the prenatal period, we had few participants in the sample who experienced many shocks, partly due to the short timeframe. Although we assessed household earnings through detailed interviews, our monthly earnings measure does not account for within-month volatility. Future work using methods such as ecological momentary assessment could capture finer-grained detail about income during pregnancy to further explore income dynamics and child outcomes. Finally, our measure of income instability used one metric of income change and focused on household earned income. Although our measures are commonly used and earning is a substantial proportion of most households' income (Morrissey et al. 2020), future research should investigate other sources of income and their relationship to instability during pregnancy.

Our findings suggest that reducing income losses during the perinatal period could be a promising intervention target. Paid family and medical leave programs are effective at reducing income loss and improving stability in the early postnatal period (Stanczyk 2019). Research investigating these programs finds benefits for parents (Aitken et al. 2015, Bilgrami et al. 2020, Bullinger 2019) and children (Bullinger 2019, Lichtman-Sadot and Bell 2017, Nandi et al. 2018, Ruhm 2000). However, parental leave support is mostly provided during the early postnatal period. Our data suggest that income instability during pregnancy could be targeted by interventions designed to support families. Pregnancy can interfere with workforce participation, especially for those with medical complications or physically demanding jobs. Low-income families, who experience more income instability on average (Wolf et al. 2014), may be especially vulnerable during this sensitive period.

Overall, our findings suggest income instability, specifically substantial income losses, is a stressor that may affect two generations. The perinatal period is vulnerable to economic instability (Stanczyk 2020) and stress's influence on the brain. Early differences in brain structure may influence developmental outcomes and pose risks for psychopathology and later cognitive functioning (Gilmore et al. 2018). Therefore, promoting economic stability during pregnancy may reduce health disparities and improve outcomes for all families.

## Acknowledgments

This work was supported by the National Institute of Health (R01HD090068; R21DA046556) and NARSAD Independent Investigator Grant. The authors thank the families that participated in the study and the individuals who supported recruitment. The authors also wish to acknowledge Brian Bello, Christian Capistrano, Ximena Calderon, Madeline Caruso, Nikisha Charles-Stazzone, Jenna Chin, Jamie Cross, Andrew Erhart, Leah Grande, Melissa Hansen, Isabella Jaramillo, Lydia Mathis, Jacqueline Martinez, Aviva Olsavsky, Rachael Ruff, Erin Schall, Rebekah Tribble, Shannon Powers, and Yun Xie for research assistance.

## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## References

- Aitken, Z., C. C. Garrett, B. Hewitt, L. Keogh, J. S. Hocking, and A. M. Kavanagh. 2015. "The Maternal Health Outcomes of Paid Maternity Leave: A Systematic Review." *Social Science & Medicine* 130: 32–41.
- Aronsson, M., K. Fuxe, Y. Dong, L. F. Agnati, S. Okret, and J. A. Gustafsson. 1988. "Localization of Glucocorticoid Receptor mRNA in the Male Rat Brain by In Situ Hybridization." *Proceedings of the National Academy of Sciences* 85, no. 23: 9331–9335.
- Avants, B. B., N. Tustison, and G. Song. 2009. "Advanced Normalization Tools (ANTS)." *JCI Insight* 2, no. 365: 1–35.
- Badihian, N., S. S. Daniali, and R. Kelishadi. 2020. "Transcriptional and Epigenetic Changes of Brain Derived Neurotrophic Factor Following Prenatal Stress: A Systematic Review of Animal Studies." *Neuroscience & Biobehavioral Reviews* 117: 211–231.
- Bania, N., and L. Leete. 2009. "Monthly Household Income Volatility in the US, 1991/92 vs. 2002/03." *Economics Bulletin* 29, no. 3: 2100–2112.
- Belli, R. F. 1998. "The Structure of Autobiographical Memory and the Event History Calendar: Potential Improvements in the Quality of Retrospective Reports in Surveys." *Memory (Hove, England)* 6, no. 4: 383–406.
- Belli, R. F., E. H. Lee, F. P. Stafford, and C. H. Chou. 2004. "Calendar and Question-List Survey Methods: Association Between Interviewer Behaviors and Data Quality." *Journal of Official Statistics* 20, no. 2: 185.
- Belsky, J., G. L. Schlomer, and B. J. Ellis. 2012. "Beyond Cumulative Risk: Distinguishing Harshness and Unpredictability as Determinants of Parenting and Early Life History Strategy." *Developmental Psychology* 48, no. 3: 662–673. <https://doi.org/10.1037/a0024454>.
- Berman, I. S., K. A. McLaughlin, N. Tottenham, et al. 2022. "Measuring Early Life Adversity: A Dimensional Approach." *Development and Psychopathology* 34, no. 2: 499–511.
- Betancourt, L. M., B. Avants, M. J. Farah, et al. 2016. "Effect of Socioeconomic Status (SES) Disparity on Neural Development in Female African-American Infants at Age 1 Month." *Developmental Science* 19, no. 6: 947–956. <https://doi.org/10.1111/desc.12344>.
- Bilgrami, A., K. Sinha, and H. Cutler. 2020. "The Impact of Introducing a National Scheme for Paid Parental Leave on Maternal Mental Health Outcomes." *Health Economics* 29, no. 12: 1657–1681.
- Bogoch, Y., Y. N. Biala, M. Linial, and M. Weinstock. 2007. "Anxiety Induced by Prenatal Stress Is Associated With Suppression of Hippocampal Genes Involved in Synaptic Function." *Journal of Neurochemistry* 101, no. 4: 1018–1030.

- Bullinger, L. R. 2019. "The Effect of Paid Family Leave on Infant and Parental Health in the United States." *Journal of Health Economics* 66: 101–116.
- Buss, C., E. P. Davis, B. Shahbaba, J. C. Pruessner, K. Head, and C. A. Sandman. 2012. "Maternal Cortisol Over the Course of Pregnancy and Subsequent Child Amygdala and Hippocampus Volumes and Affective Problems." *Proceedings of the National Academy of Sciences* 109, no. 20: 1312–1319.
- Buthmann, J. L., C. Antonacci, J. P. Uy, L. R. Borchers, J. G. Miller, and I. H. Gotlib. 2025. "Stress Reactivity Moderates the Association Between Early Experiences of Unpredictability and Emotional Problems in Adolescents." *Neurobiology of Stress* 34: 100706. <https://doi.org/10.1016/j.ynstr.2024.100706>.
- Charil, A., D. P. Laplante, C. Vaillancourt, and S. King. 2010. "Prenatal Stress and Brain Development." *Brain Research Reviews* 65, no. 1: 56–79.
- Cheng, S., K. Kosidou, B. Burström, C. Björkenstam, A. R. Pebley, and E. Björkenstam. 2020. "Precarious Childhoods: Childhood Family Income Volatility and Mental Health in Early Adulthood." *Social Forces* 99, no. 2: 672–699.
- Cherel, M., F. Budin, M. Prastawa, et al. 2015. "Automatic Tissue Segmentation of Neonate Brain MR Images With Subject-Specific Atlases." *SPIE Medical Imaging*.
- Cooper, M., and A. J. Pugh. 2020. "Families Across the Income Spectrum: A Decade in Review." *Journal of Marriage and Family* 82: 272–299. <https://doi.org/10.1111/jomf.12623>.
- Cox, J. L., J. M. Holden, and R. Sagovsky. 1987. "Detection of Postnatal Depression. Development of the 10-Item Edinburgh Postnatal Depression Scale." *British Journal of Psychiatry: The Journal of Mental Science* 150: 782–786.
- De France, K., G. W. Evans, G. H. Brody, and S. N. Doan. 2022. "Cost of Resilience: Childhood Poverty, Mental Health, and Chronic Physiological Stress." *Psychoneuroendocrinology* 144: 105872. <https://doi.org/10.1016/j.psyneuen.2022.105872>.
- DiTosto, J. D., K. Holder, E. Soyemi, M. Beestrum, and L. M. Yee. 2021. "Housing Instability and Adverse Perinatal Outcomes: A Systematic Review." *American Journal of Obstetrics & Gynecology MFM* 3, no. 6: 100477.
- Dufford, A. J., and P. Kim. 2017. "Family Income, Cumulative Risk Exposure, and White Matter Structure in Middle Childhood." *Frontiers in Human Neuroscience* 11: 547. <https://doi.org/10.3389/fnhum.2017.00547>.
- Elfassy, T., S. L. Swift, M. M. Glymour, et al. 2019. "Associations of Income Volatility With Incident Cardiovascular Disease and All-Cause Mortality in a US Cohort." *Circulation* 139, no. 7: 850–859. <https://doi.org/10.1161/CIRCULATIONAHA.118.035521>.
- Ellis, B. J., M. A. Sheridan, J. Belsky, and K. A. McLaughlin. 2022. "Why and How Does Early Adversity Influence Development? Toward an Integrated Model of Dimensions of Environmental Experience." *Development and Psychopathology* 34, no. 2: 447–471.
- Evans, G. W., M. J. Farah, and D. A. Hackman. 2021. "Early Childhood Poverty and Adult Executive Functioning: Distinct, Mediating Pathways for Different Domains of Executive Functioning." *Developmental Science* 24, no. 5: 13084.
- Evans, G. W., and P. Kim. 2013. "Childhood Poverty, Chronic Stress, Self-Regulation, and Coping." *Child Development Perspectives* 7, no. 1: 43–48. <https://doi.org/10.1111/cdep.12013>.
- Farah, M. J. 2017. "The Neuroscience of Socioeconomic Status: Correlates, Causes, and Consequences." *Neuron* 96, no. 1: 56–71. <https://doi.org/10.1016/j.neuron.2017.08.034>.
- Faul, F., E. Erdfelder, A.-G. Lang, and A. Buchner. 2007. "G\*Power 3: A Flexible Statistical Power Analysis Program for the Social, Behavioral, and Biomedical Sciences." *Behavior Research Methods* 39, no. 2: 175–191. <https://doi.org/10.3758/BF03193146>.
- Fonov, V., A. C. Evans, K. Botteron, et al. 2011. "Unbiased Average Age-Appropriate Atlases for Pediatric Studies." *Neuroimage* 54, no. 1: 313–327.
- Frankiensztajn, L. M., E. Elliott, and O. Koren. 2020. "The Microbiota and the Hypothalamus-Pituitary-Adrenocortical (HPA) Axis, Implications for Anxiety and Stress Disorders." *Current Opinion in Neurobiology* 62: 76–82.
- Freide, E., and M. Weinstock. 1984. "The Effects of Prenatal Exposure to Predictable or Unpredictable Stress on Early Development in the Rat." *Developmental Psychobiology: The Journal of the International Society for Developmental Psychobiology* 17, no. 6: 651–660.
- Gao, W., S. Alcauter, A. Elton, et al. 2015. "Functional Network Development During the First Year: Relative Sequence and Socioeconomic Correlations." *Cerebral Cortex* 25, no. 9: 2919–2928. <https://doi.org/10.1093/cercor/bhu088>.
- Gennetian, L. A., C. Rodrigues, H. D. Hill, and P. A. Morris. 2018. "Stability of Income and School Attendance Among NYC Students of Low-Income Families." *Economics of Education Review* 63: 20–30.
- Gennetian, L. A., and E. Shafir. 2015. "The Persistence of Poverty in the Context of Financial Instability: A Behavioral Perspective." *Journal of Policy Analysis and Management* 34, no. 4: 904–936. <https://doi.org/10.1002/pam.21854>.
- Gilmore, J. H., R. C. Knickmeyer, and W. Gao. 2018. "Imaging Structural and Functional Brain Development in Early Childhood." *Nature Reviews Neuroscience* 19, no. 3: 123–137.
- Glasner, T., and W. van der Vaart. 2009. "Applications of Calendar Instruments in Social Surveys: A Review." *Quality and Quantity* 43: 333–349.
- Grasset, L., M. M. Glymour, T. Elfassy, et al. 2019. "Relation Between 20-Year Income Volatility and Brain Health in Midlife: The CARDIA Study." *Neurology* 93, no. 20: 1890–1899.
- Gross, R. S., A. L. Mendelsohn, M. M. Arana, and M. J. Messito. 2019. "Food Insecurity During Pregnancy and Breastfeeding by Low-Income Hispanic Mothers." *Pediatrics* 143, no. 6: e20184113. <https://doi.org/10.1542/peds.2018-4113>.
- Haile, C. N., T. GrandPre, and T. A. Kosten. 2001. "Chronic Unpredictable Stress, but Not Chronic Predictable Stress, Enhances the Sensitivity to the Behavioral Effects of Cocaine in Rats." *Psychopharmacology* 154: 213–220.
- Hair, N. L., J. L. Hanson, B. L. Wolfe, and S. D. Pollak. 2015. "Association of Child Poverty, Brain Development, and Academic Achievement." *JAMA Pediatrics* 169, no. 9: 822–829. <https://doi.org/10.1001/jamapediatrics.2015.1475>.
- Hardy, B., and J. P. Ziliak. 2014. "Decomposing Trends in Income Volatility: The 'Wild Ride' at the Top and Bottom." *Economic Inquiry* 52: 459–476. <https://doi.org/10.1111/ecin.12044>.
- Hardy, B. L. 2014. "Childhood Income Volatility and Adult Outcomes." *Demography* 51, no. 5: 1641–1665.
- Hardy, B. L. 2017. "Income Instability and the Response of the Safety Net." *Contemporary Economic Policy* 35, no. 2: 312–330. <https://doi.org/10.1111/coep.12187>.
- Havranek, M. M., B. Bolliger, S. Roos, C. R. Pryce, B. B. Quednow, and E. Seifritz. 2016. "Uncontrollable and Unpredictable Stress Interacts With Subclinical Depression and Anxiety Scores in Determining Anxiety Response." *Stress (Amsterdam, Netherlands)* 19, no. 1: 53–62. <https://doi.org/10.3109/10253890.2015.1117449>.
- Heflin, C., T. W. Morrissey, and H. Kim. 2024. *Do the Temporary Assistance for Needy Families and the Supplemental Nutrition Assistance Programs Function as Parental Leave for Low-Income Women?* Working Paper. Syracuse University.
- Hill, H. D., P. Morris, L. A. Gennetian, S. Wolf, and C. Tubbs. 2013. "The Consequences of Income Instability for Children's Well-Being." *Child Development Perspectives* 7, no. 2: 85–90. <https://doi.org/10.1111/cdep.12018>.



- Hu, C., Y. Luo, H. Wang, et al. 2017. "Re-Evaluation of the Interrelationships Among the Behavioral Tests in Rats Exposed to Chronic Unpredictable Mild Stress." *PLoS ONE* 12, no. 9: e0185129. <https://doi.org/10.1371/journal.pone.0185129>.
- Humphreys, K. L., M. C. Camacho, M. C. Roth, and E. C. Estes. 2020. "Prenatal Stress Exposure and Multimodal Assessment of Amygdala-Medial Prefrontal Cortex Connectivity in Infants." *Developmental Cognitive Neuroscience* 46: 100877.
- Johnson, S. B., J. L. Riis, and K. G. Noble. 2016. "State of the Art Review: Poverty and the Developing Brain." *Pediatrics* 137, no. 4: e20153075. <https://doi.org/10.1542/peds.2015-3075>.
- Jones, S. L., R. Dufoix, D. P. Laplante, et al. 2019. "Larger Amygdala Volume Mediates the Association Between Prenatal Maternal Stress and Higher Levels of Externalizing Behaviors: Sex Specific Effects in Project Ice Storm." *Frontiers in Human Neuroscience* 13: 144.
- Kim, P., H. Chen, A. J. Dufford, R. Tribble, J. Gilmore, and W. Gao. 2021. *Intergenerational Neuroimaging Study: Mother-Infant Functional Connectivity Similarity and the Role of Infant and Maternal Factors*. Cerebral Cortex.
- Kim, P., G. W. Evans, M. Angstadt, et al. 2013. "Effects of Childhood Poverty and Chronic Stress on Emotion Regulatory Brain Function in Adulthood." *Proceedings of the National Academy of Sciences* 110, no. 46: 18442–18447. <https://doi.org/10.1073/pnas.1308240110>.
- Kim, P., G. W. Evans, E. Chen, G. Miller, and T. Seeman. 2018. "How Socioeconomic Disadvantages Get Under the Skin and Into the Brain to Influence Health Development Across the Lifespan." In *Handbook of Life Course Health Development*, Springer, 463–497.
- Kim, P., R. Tribble, A. K. Olsavsky, et al. 2020. "Associations Between Stress Exposure and New Mothers' brain Responses to Infant Cry Sounds." *Neuroimage* 223: 117360.
- Lautarescu, A., M. C. Craig, and V. Glover. 2020. "Prenatal Stress: Effects on Fetal and Child Brain Development." *International Review of Neurobiology* 150: 17–40.
- Lautarescu, A., D. Pecheva, C. Nosarti, et al. 2020. "Maternal Prenatal Stress Is Associated With Altered Uncinate Fasciculus Microstructure in Premature Neonates." *Biological Psychiatry* 87, no. 6: 559–569.
- Leijser, L. M., A. Siddiqi, and S. P. Miller. 2018. "Imaging Evidence of the Effect of Socio-Economic Status on Brain Structure and Development." *Seminars in Pediatric Neurology* 27: 26–34. <https://doi.org/10.1016/j.spen.2018.03.004>.
- Li, X., M. N. I. Qureshi, D. P. Laplante, et al. 2023. "Neural Correlates of Disaster-Related Prenatal Maternal Stress in Young Adults From Project Ice Storm: Focus on Amygdala, Hippocampus, and Prefrontal Cortex." *Frontiers in Human Neuroscience* 17: 1094039.
- Lichtman-Sadot, S., and N. P. Bell. 2017. "Child Health in Elementary School Following California's Paid Family Leave Program." *Journal of Policy Analysis and Management: [The Journal of the Association for Public Policy Analysis and Management]* 36, no. 4: 790–827. <https://doi.org/10.1002/pam.22012>.
- Lu, Y. C., K. Kapse, N. Andersen, et al. 2021. "Association Between Socioeconomic Status and In Utero Fetal Brain Development." *JAMA Network Open* 4, no. 3: 213526–213526.
- Mancuso, R. A., K. M. Ross, E. Accortt, et al. 2024. "Prenatal Mood and Anxiety Disorders and Associated Cytokine Changes." *Journal of Affective Disorders* 347: 635–644.
- Marchisella, F., K. C. Creutzberg, V. Begni, et al. 2021. "Exposure to Prenatal Stress Is Associated With an Excitatory/Inhibitory Imbalance in Rat Prefrontal Cortex and Amygdala and an Increased Risk for Emotional Dysregulation." *Frontiers in Cell and Developmental Biology* 9: 653 384.
- Marcynyszyn, L. A., G. W. Evans, and J. Eckenrode. 2008. "Family Instability During Early and Middle Adolescence." *Journal of Applied Developmental Psychology* 29, no. 5: 380–392.
- Mareckova, K., R. Marecek, L. Andryskova, M. Brazdil, and Y. S. Nikolova. 2022. "Impact of Prenatal Stress on Amygdala Anatomy in Young Adulthood: Timing and Location Matter." *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* 7, no. 2: 231–238.
- McEwen, B. S., E. A. Gould, and R. R. Sakai. 1992. "The Vulnerability of the Hippocampus to Protective and Destructive Effects of Glucocorticoids in Relation to Stress." *British Journal of Psychiatry* 160, no. S15: 18–23.
- McLaughlin, K. A., and M. A. Sheridan. 2016. "Beyond Cumulative Risk: A Dimensional Approach to Childhood Adversity." *Current Directions in Psychological Science* 25, no. 4: 239–245. <https://doi.org/10.1177/0963721416655883>.
- McLaughlin, K. A., M. A. Sheridan, K. L. Humphreys, J. Belsky, and B. J. Ellis. 2021. "The Value of Dimensional Models of Early Experience: Thinking Clearly About Concepts and Categories." *Perspectives on Psychological Science* 16, no. 6: 1463–1472. <https://doi.org/10.1177/1745691621992346>.
- McLaughlin, K. A., M. A. Sheridan, and H. K. Lambert. 2014. "Childhood Adversity and Neural Development: Deprivation and Threat as Distinct Dimensions of Early Experience." *Neuroscience and Biobehavioral Reviews* 47: 578–591. <https://doi.org/10.1016/j.neubiorev.2014.10.012>.
- Miller, P., K. Whitfield, L. Betancur, and E. Votruba-Drzal. 2021. "Income Dynamics and Behavior Problems in Early Childhood, Middle Childhood, and the Transition to Adolescence." *Journal of Applied Developmental Psychology* 77: 101345.
- Monteiro, S., S. Roque, D. de Sá-Calçada, N. Sousa, M. Correia-Neves, and J. J. Cerqueira. 2015. "An Efficient Chronic Unpredictable Stress Protocol to Induce Stress-Related Responses in C57BL/6 Mice." *Frontiers in Psychiatry* 6: 6. <https://doi.org/10.3389/fpsy.2015.00006>.
- Moog, N. K., S. Entringer, J. M. Rasmussen, et al. 2018. "Intergenerational Effect of Maternal Exposure to Childhood Maltreatment on Newborn Brain Anatomy." *Biological Psychiatry* 83, no. 2: 120–127.
- Moog, N. K., S. Nolvi, T. S. Kleih, et al. 2021. "Prospective Association of Maternal Psychosocial Stress in Pregnancy With Newborn Hippocampal Volume and Implications for Infant Social-Emotional Development." *Neurobiology of Stress* 15: 100368. <https://doi.org/10.1016/j.ynstr.2021.100368>.
- Morrissey, T. W., Y. Cha, S. Wolf, and M. Khan. 2020. "Household Economic Instability: Constructs, Measurement, and Implications." *Children and Youth Services Review* 118: 105502.
- Nandi, A., D. Jahagirdar, M. C. Dimitris, et al. 2018. "The Impact of Parental and Medical Leave Policies on Socioeconomic and Health Outcomes in OECD Countries: A Systematic Review of the Empirical Literature." *Milbank Quarterly* 96, no. 3: 434–471. <https://doi.org/10.1111/1468-0009.12340>.
- Noble, K. G., S. M. Houston, N. H. Brito, et al. 2015. "Family Income, Parental Education and Brain Structure in Children and Adolescents." *Nature Neuroscience* 18, no. 5: 773–778. <https://doi.org/10.1038/nn.3983>.
- O'Donnell, K., T. G. O'Connor, and V. Glover. 2009. "Prenatal Stress and Neurodevelopment of the Child: Focus on the HPA Axis and Role of the Placenta." *Developmental Neuroscience* 31, no. 4: 285–292. <https://doi.org/10.1159/000216539>.
- Overfeld, J., S. Entringer, J. M. Rasmussen, et al. 2020. "Neonatal Hippocampal Volume Moderates the Effects of Early Postnatal Enrichment on Cognitive Development." *Developmental Cognitive Neuroscience* 45: 100820. <https://doi.org/10.1016/j.dcn.2020.100820>.
- Peetz, J., J. Robson, and S. Xuereb. 2021. "The Role of Income Volatility and Perceived Locus of Control in Financial Planning Decisions." *Frontiers in Psychology* 12: 1818. <https://doi.org/10.3389/fpsyg.2021.638043>.
- Posner, J., J. Cha, A. K. Roy, et al. 2016. "Alterations in Amygdala-Prefrontal Circuits in Infants Exposed to Prenatal Maternal Depression." *Translational Psychiatry* 6, no. 11: 935–935.
- Prause, J., D. Dooley, and J. Huh. 2009. "Income Volatility and Psychological Depression." *American Journal of Community Psychology* 43: 57–70. <https://doi.org/10.1007/s10464-008-9219-3>.

- Qiu, A., A. Rifkin-Graboi, H. Chen, et al. 2013. "Maternal Anxiety and Infants' hippocampal Development: Timing Matters." *Translational Psychiatry* 3, no. 9: 306–306.
- Rakesh, D., and S. Whittle. 2021. "Socioeconomic Status and the Developing Brain—A Systematic Review of Neuroimaging Findings in Youth." *Neuroscience and Biobehavioral Reviews* 130: 379–407. <https://doi.org/10.1016/j.neubiorev.2021.08.027>.
- Ramphal, B., D. J. Whalen, J. K. Kenley, et al. 2020. "Brain Connectivity and Socioeconomic Status at Birth and Externalizing Symptoms at Age 2 Years." *Developmental Cognitive Neuroscience* 45: 100811. <https://doi.org/10.1016/j.dcn.2020.100811>.
- Rifkin-Graboi, A., J. Bai, H. Chen, et al. 2013. "Prenatal Maternal Depression Associates With Microstructure of Right Amygdala in Neonates at Birth." *Biological Psychiatry* 74, no. 11: 837–844.
- Roshan-Milani, S., B. Seyyedabadi, E. Saboory, N. Parsamanesh, and N. Mehranfar. 2021. "Prenatal Stress and Increased Susceptibility to Anxiety-Like Behaviors: Role of Neuroinflammation and Balance Between GABAergic and Glutamatergic Transmission." *Stress (Amsterdam, Netherlands)* 24, no. 5: 481–495.
- Ruhm, C. J. 2000. "Parental Leave and Child Health." *Journal of Health Economics* 19, no. 6: 931–960. [https://doi.org/10.1016/S0167-6296\(00\)00047-3](https://doi.org/10.1016/S0167-6296(00)00047-3).
- Scheinost, D., S. H. Kwon, C. Lacadie, et al. 2016. "Prenatal Stress Alters Amygdala Functional Connectivity in Preterm Neonates." *NeuroImage: Clinical* 12: 381–388.
- Scheinost, D., M. N. Spann, L. McDonough, B. S. Peterson, and C. Monk. 2020. "Associations Between Different Dimensions of Prenatal Distress, Neonatal Hippocampal Connectivity, and Infant Memory." *Neuropsychopharmacology* 45, no. 8: 1272–1279.
- Shrider, E. A., M. Kollar, F. Chen, and J. Semega. n.d. *Income and Poverty in the United States: 2020*. Census.Gov. Retrieved January 5, 2025, From <https://www.census.gov/library/publications/2021/demo/p60-273.html>.
- Spann, M. N., R. Bansal, X. Hao, T. S. Rosen, and B. S. Peterson. 2020. "Prenatal Socioeconomic Status and Social Support Are Associated With Neonatal Brain Morphology, Toddler Language and Psychiatric Symptoms." *Child Neuropsychology* 26, no. 2: 170–188.
- Spielberger, C. D., R. L. Gorsuch, R. Lushene, P. R. Vagg, and G. A. Jacobs. 1983. *Manual for the State-Trait Anxiety Inventory*. Consulting Psychologists Press.
- Stanczyk, A. B. 2019. "Does Paid Family Leave Improve Household Economic Security Following a Birth? Evidence From California." *Social Service Review* 93, no. 2: 262–304.
- Stanczyk, A. B. 2020. "The Dynamics of US Household Economic Circumstances Around a Birth." *Demography* 57, no. 4: 1271–1296.
- Triplett, R. L., R. E. Lean, A. Parikh, et al. 2022. "Association of Prenatal Exposure to Early-Life Adversity With Neonatal Brain Volumes at Birth." *JAMA Network Open* 5, no. 4: 227045–227045.
- Tustison, N. J., B. B. Avants, P. A. Cook, et al. 2010. "N4ITK: Improved N3 Bias Correction." *IEEE Transactions on Medical Imaging* 29, no. 6: 1310–1320.
- Tustison, N. J., P. A. Cook, A. J. Holbrook, et al. 2021. "The ANTsX Ecosystem for Quantitative Biological and Medical Imaging." *Scientific Reports* 11, no. 1: 9068.
- Usacheva, M., D. Choe, S. Liu, S. Timmer, and J. Belsky. 2022. "Testing the Empirical Integration of Threat-Deprivation and Harshness-Unpredictability Dimensional Models of Adversity." *Development and Psychopathology* 34, no. 2: 513–526.
- Wen, D. J., J. S. Poh, S. N. Ni, et al. 2017. "Influences of Prenatal and Postnatal Maternal Depression on Amygdala Volume and Microstructure in Young Children." *Translational Psychiatry* 7, no. 4: 1103–1103.
- Western, B., D. Bloome, B. Sosnaud, and L. M. Tach. 2016. "Trends in Income Insecurity Among US Children, 1984–2010." *Demography* 53, no. 2: 419–447.
- Whitfield, K., L. Betancur, P. Miller, and E. Votruba-Drzal. 2021. "Longitudinal Links Between Income Dynamics and Young Adult Socioeconomic and Behavioral Health Outcomes." *Youth & Society* 53, no. 7: 1181–1210. <https://doi.org/10.1177/0044118X21996382>.
- Wolf, S., L. A. Gennetian, P. A. Morris, and H. D. Hill. 2014. "Patterns of Income Instability Among Low-and Middle-Income Households With Children." *Family Relations* 63, no. 3: 397–410.
- Wolf, S., and T. W. Morrissey. 2017. "Economic Instability, Food Insecurity, and Child Health During and After the Great Recession." *Social Service Review* 91, no. 3: 534–570.
- Yushkevich, P. A., J. Piven, H. C. Hazlett, et al. 2006. "User-Guided 3D Active Contour Segmentation of Anatomical Structures: Significantly Improved Efficiency and Reliability." *Neuroimage* 31, no. 3: 1116–1128.
- Zach, P., J. Mrzilková, L. Řezáčová, A. Stuchlík, and K. Valeš. 2010. "Delayed Effects of Elevated Corticosterone Level on Volume of Hippocampal Formation in Laboratory Rat." *Physiological Research* 59, no. 6: 985–996. <https://doi.org/10.33549/physiolres.931904>.
- Zachrisson, H. D., and E. Dearing. 2015. "Family Income Dynamics, Early Childhood Education and Care, and Early Child Behavior Problems in Norway." *Child Development* 86: 425–440. <https://doi.org/10.1111/cdev.12306>.
- Zimmer, C., C. C. Taff, D. R. Ardia, et al. 2020. "Environmental Unpredictability Shapes Glucocorticoid Regulation Across Populations of Tree Swallows." *Scientific Reports* 10, no. 1: 13682. <https://doi.org/10.1038/s41598-020-70161-4>.

## Supporting Information

Additional supporting information can be found online in the Supporting Information section.